Lab 2: Hardware Interfacing (Motor, Encoder), Filtering, and Modeling (Differential Equation, Block Diagram, Transfer Function)

Objectives: In this lab, you will work with a DC motor (actuator) and a rotary encoder (sensor). You will model the DC motor using three approaches: differential equations, block diagrams, and transfer functions. You will compare the behavior of these models with that of a physical DC motor. You will perform encoder signal processing to convert raw encoder counts into shaft angle and angular speed, and implement a first-order low-pass filter to reduce measurement noise.

Timeline

This lab spans two weeks. Complete the pre-lab tasks **before the first lab session**. Review all reading materials **during the two weeks but before attending each lab session**. Do not wait until the lab sessions to complete these preparations.

Pre-Lab Tasks

Due: Before the first lab session. Note: You are not required to upload anything to D2L. This pre-lab is for your own practice and preparation only.

Complete the following tasks:

- MATLAB Course to take: Control System Modeling Essentials (about an hour). Please note that Simscape is out of the scope of this course.
  `https://matlabacademy.mathworks.com/details/control-system-modeling-essentials/otmlslcsme`

- **Draft Section 6.5 before the first session.** Derive the differential equations for the DC motor and draft the `.mlx` Live Script that solves them numerically. This does not require the QUBE-Servo 2 hardware, and the scheduled lab period does not allow enough time to complete the whole lab if it is left until then.
  Reuse the Forward Euler function you wrote in Lab 1 — do not write a new solver.

Required Reading

Complete during the two-week lab period, before attending lab sessions.

From the Reference Book, *Feedback Control of Dynamic Systems* (8th edition), review the following topics:

- Transfer function: Examples 2.1 and 2.2.
- DC rotary servo motor modeling: Section 2.3.2.
- Block Diagram: Sections 1.1, 1.2, 3.2.1, 3.2.2.
- Laplace Transform: Sections 3.1.3, 3.1.4, 3.1.5, 3.1.6, 3.1.7.

Read the following materials on D2L:

- Lab 2 Manual
- Lab 2 Presentation
- Lecture Slides
- Position and Displacement Sensors – 10 pages – From Chapter 13, Book: *Introduction to Mechatronic Design*, by J. Edward Carryer, R. Matthew Ohline, and Thomas W. Kenny. This covers potentiometers, optical encoders, quadrature encoder, reflective infrared sensors, and ultrasonic displacement sensors.
- Sensors – 12 pages – From Chapter 7, Book *Fundamentals of Mechatronics*, by M. Jouanh. This covers sensor performance, displacement measurement, potentiometers, incremental encoder, and absolute encoder.